

Reproduction of sthalles/SimCLR: SimCLR ResNet-18 on CIFAR-10 with linear evaluation

K-Veritas Team

Submitted by K-Veritas Team

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Independent reproduction · Artifact evaluation · Computer vision

Abstract

Pretrained a ResNet-18 SimCLR model on CIFAR-10 with run.py (50 epochs, batch 256, Adam lr 3e-4, temperature 0.07, fp16 mixed precision, the script's fixed seed 0), then ran the repo's linear evaluation from feature_eval/mini_batch_logistic_regression_evaluator.ipynb as a script: frozen backbone, logistic-regression head trained with Adam for 100 epochs on CIFAR-10, top-1/top-5 test accuracy each epoch. The notebook code is unchanged except that it loads the checkpoint just trained instead of downloading a published one; run.py gained one print per epoch of the values it already logs. CIFAR-10 downloaded by torchvision. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	SimCLR CIFAR-10 linear eval	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: SimCLR CIFAR-10 linear eval

Run	Metric	Final value	Step
1	contrastive_loss	2.46374	49
1	contrastive_top1	50	49
1	linear_train_top1	66.0738	99
1	linear_test_top1	63.0314	99

Run	Inline claim	Value	Phase
1	linear_eval_test_top1	63.0314	
1	linear_eval_test_top5	96.3971	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	1239.0 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	cb855eaa0be488f58ae2c4502d26287676e6578a31464823179db0888716d22e	7.8 MB
r1.kvbundle.zip	719c811c6120a0fcdda61c9028456c513c84eb9ce99ffc9e87022e095d0d52c1	21 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2002.05709>

Reproduction of: <https://github.com/sthalles/SimCLR/tree/1848fc934ad844ae630e6c452300433fe99acfd9>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: simclr, self-supervised, contrastive-learning, cifar-10.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (SimCLR CIFAR-10 linear eval), kv:2610.00007v1/r1. Data hash f549474966292babd57d12e269d8acc3b6077e6c111df2ade35d5815ede9597e.
<https://kveritas.org/records/2610.00007v1/r1.pdf>